Foundation

STATISTICS - 2

WEEK-1

Agenda:

1. Joint Distribution
2. Marginal Distribution
3. Conditional Distribution
4. Independence
5. Covariance & Correlation

① Joint PMF

Probability Mass Function

$$f_{X,Y}(x,y) = P(X = x \text{ and } Y = y)$$

$0 \le f(x,y) \le 1$

t_2 / t_1 (Y \ X)	0	1
0	1/4	1/4
1	1/4	1/4

Sum of all entries = 1

f → represent Cumulative prob.

Q.1 Dice Problem → experiment consists of rolling unbiased die 2 times

Calculate $f_{X_1X_2}(3,2)$

$X_i \sim \text{Uniform}\{1,2,3,4,5,6\}$

$P(X_1 = 3 \text{ and } X_2 = 2)$

$= \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$

Q.2. The joint PMF of 2 discrete variables x and y given in Table, which of the following are possible value of 'a' and b;

Hint: Sum of all entries = 1

$a + b = \frac{1}{2}$ → find best on

Q.3. The joint PMF of 2 discrete random variables x and y is given by

$$g_{XY}(x,y) = \frac{xy}{9}, \quad x, y \in \{1,2\}.$$

$P(x+y=1) \to 0$

$P(x-y=1) = \frac{2}{9}$

$x = 2$

$y = 1$

$g_{XY}(2,1) = \frac{xy}{9} = \frac{(2)(1)}{9} = \frac{2}{9}$

X	Y	X+Y	X-Y
1	1	2	0
2	2	4	0
1	2	3	-1
2	1	3	1

Marginal PMF [Sum of rows / colums of joint table]

- $f_X(x) = \sum_y f_{X,Y}(x,y)$
- $f_Y(y) = \sum_x f_{X,Y}(x,y)$

Q.1 The joint PMF of 2 discrete random var. x and y is given by $f_{xy}(x,y) = \frac{xy}{4}$ $x, y \in \{1, 2\}$

calculate $f_x(1) + f_x(2) = 1$

$1/3 + 2/3$

Sol.

$$f_x(1) = f(1,1) + f(1,2) = \frac{1}{9} + \frac{2}{9} = \frac{3}{9} = \frac{1}{3}$$

$$f_x(2) = f(2,1) + f(2,2) = \frac{2}{9} + \frac{4}{9} = \frac{2}{3}$$

Conditional Distribution

$$P(X = x \mid Y = y) = \frac{P(X = x \text{ and } Y = y)}{P(Y = y)}$$

Distributions:

Uniformal
Normal
Exponential
in week-4

(1) Binomial Distribution $X \sim \text{Binomia}(n, p)$

$$\left[P(X = x) = {}^{n}C_{x}\, p^{x} q^{n-x} \right] \quad n = 0, 1, 2 \ldots$$

(2) Poisson Distribution $X \sim \text{Poisson}(\lambda)$

$$\left[P(X = x) = \frac{e^{-\lambda} (\lambda)^{x}}{x!} \right] \quad n = 0, 1, 2 \ldots$$

(3) Bernoulli Distribution $X \sim \text{Bernoulli}$

$$\left[P(X = x) = p^{x} (1-p)^{1-x} \right] \quad x \sim \{0, 1\}$$

(4) Geometric Distribution $X \sim \text{Geometric}$

$$\left[P(X = x) = (1-p)^{x} \cdot p \right] \quad x = \{1, 2, 3, \ldots\}$$

$$p \cdot (1-p)^{k-1}$$

~~# Adv. Ques.~~

- Joint PMF of Multiple Variables

$$\hookrightarrow f_{X_1 X_2}(x_1, x_2) = \sum_{x_3} f_{X_1 X_2 X_3}(x_1, x_2, x_3)$$

- Conditioning with Multiple Variables

$$\hookrightarrow f_{X_1 \mid X_2}(x_1 \mid x_2) = \frac{f_{X_1 X_2}(x_1, x_2)}{f_{X_2}(x_2)}$$

Q.1

t_1	t_2	t_3	t_4	$f_{X_1X_2X_3X_4}(t_1, t_2, t_3, t_4)$
0	0	0	0	1/8
0	1	0	0	1/8
0	1	1	0	1/8
0	0	0	1	1/8
1	0	1	1	1/8
1	1	1	1	1/8
1	1	1	0	1/8
1	1	1	1	1/8

$$P(X_3 = 1, X_4 = 1 \mid X_1 = 1, X_2 = 1).$$

$$\hookrightarrow \frac{\frac{1}{8} + \frac{1}{8}}{\frac{1}{8} + \frac{1}{8} + \frac{1}{8}} = \frac{2}{3}$$

Q.2. Let $X_1, X_2, X_3 \sim f_{X_1,X_2,X_3}$ where $T_{X_i} = \{0,1\}$, for each i.
If $f_{X_1,X_2,X_3}(0,0,1) = \frac{1}{5}$, $f_{X_3}(0) = \frac{2}{5}$, $f_{X_2|X_3=1}(0) = \frac{2}{3}$,
find $f_{X_1|X_2=0,X_3=1}(0)$.

Sol.

$$\frac{f_{X_1,X_2,X_3}(0,0,1) \rightarrow 1/5}{f_{X_2=0,X_3=1} \rightarrow 2/5} = \frac{1}{2}$$

$$\frac{f_{X_2=0,X_3=1}}{f_{X_3=1} \rightarrow \left(1-\frac{2}{5}\right) = \frac{3}{5}} = \frac{2}{3}$$

$$f_{X_2=0,X_3=1} = \frac{2}{\cancel{3}} \cdot \frac{\cancel{3}}{5} = \frac{2}{5}$$

WEEK-2

Independence of Random Variables

For random var, X and Y are independent if $\rightarrow$ $P(X=x, Y=y) = P(X=x) \cdot P(Y=y)$

for continuous var. $\rightarrow$ $f_{XY}(x,y) = f_X(x) \cdot f_Y(y)$

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i.i.d (Independent Indentically distributed)

$$P(X_1 > j, X_2 > j, \ldots, X_n > j) = [P(X > j)]^n$$

- Independent
- Same distribution

if $X \sim$ Geometric(p)

Q.1 Let $X_1, X_2, X_3, \ldots, X_{10} \sim$ i.i.d. Geometric$\left(\frac{1}{5}\right)$.
Find the Probability that $(X_1 > 10, X_2 > 10, \ldots X_{10} > 10)$.

Sol.
- Geometric Distr: $= P(X > k) = (1-p)^k$
$= (0.8)^{10}$
- Use Independence: $= P(X_1 > 10, \ldots, X_{10} > 10)$
$\Rightarrow \prod_{i=1}^{10} P(X_i > 10) = (0.8^{10})^{10} = 0.8^{100}$